

Yong CHENG

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EDUCATION

George Mason University - Computer Science, Ph.D. - RA, advised by Prof. Yotam Gingold - TA for CS262 Introduction to Low-Level Programming, CS471 Operating Systems	Fairfax, VA, US 2025 – present
Washington University in St. Louis - Computer Science, M.S. - Overall GPA 3.9/4.0 - TA for CSE 332/504 Object-Oriented Software Development Laboratory	St. Louis, MO, US 2021 - 2023
Shanghai Ocean University (Mathematics) Information and Computing Science, B.S. - Major GPA 3.9/4.0 (1st / 58) Overall GPA 3.5/4.0	Shanghai, China 2016 - 2021

PUBLICATIONS

RBF Your SDF: Radial Basis Function Interpolation of Signed Distance Fields with Implied Tangent Points Yong Cheng, Yotam Gingold. <i>ACM SIGGRAPH Asia 2026, Conference Track</i>	2026
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SKILLS & INTERESTS

Programming	C/C++, Python, Mathematica, MATLAB, JavaScript, Git
Software & OS	Blender, Inkscape, Photoshop, Linux/macOS/Windows
Academic Interests	Geometry Processing, Surface Reconstruction

WORK EXPERIENCE

NIO Inc. Algorithm Engineer High-Definition (HD) map, Autonomous Driving, C++	Shanghai, China May 2024 – May 2025
Map Data Auto-Generation: Analyzed and implemented algorithms from academic papers to compute map data matching relationships, ensuring adherence to industrial standards. Utilized matching relationships between data sources to merge disparate datasets, streamlining HD map auto-generation processes. Intersection Range Identification: Designed and developed an algorithm for the automatic detection of intersection ranges, significantly reducing short link issues, redundancy, and road disconnections.	
International Digital Economy Academy (IDEA) Software Engineer, Geometry and Graphics Algorithms CAD/CAE, BIM, C++	Shanghai, China Jun. 2023 - Mar. 2024
- Implemented a skeletonization algorithm for automatic generation of roofs. - Developed infrastructure libraries: curve and surface discretization, bounding box computation, quartic equation solver, and Quadtree implementation. - Enhanced the loop search algorithm with caching and Quadtree integration, gaining over 100× speedup, and rewrote the “loop to face” algorithm for a 1000× performance boost. - Designed and implemented efficient analytic intersection algorithms for a line with a cylinder, a cone, and a torus, ensuring high-speed computation with distance tolerance considerations. - Implemented a general geometric intersection algorithm to solve complex curve and surface intersection problems, whose precision surpasses industry standards like Parasolid even in extreme scenarios.	

Elekta Inc.

St. Louis, MO

Software Engineering Intern | Medical Image Management, JavaScript, Node.js

May 2022 - Aug. 2022

- Developed a web-based software for managing DICOM files in radiation therapy. Enhanced software capabilities by adding visualization, parsing, recursive indexing, querying, and file upload/download functionalities. Integrated a database and blob storage.

RESEARCH EXPERIENCE

Surface Reconstruction from Signed Distance Fields | C++, Python

Fairfax, VA

Advised by Prof. Yotam Gingold, George Mason University

2025 – present

- **Method:** Developed an interpolatory surface reconstruction method that treats each SDF sample as a tangent-sphere constraint, recovers implied surface points, and fits an implicit surface with a partition-of-unity RBF system. Accepted to SIGGRAPH Asia 2026.
- **Results:** Reduced Chamfer distance by 40–85% relative to the prior state of the art across sampling resolutions from 6^3 to 100^3 ; recovered sharp corners from collapsed exposed regions of the spheres.
- **Scalability & Implementation:** Kept the pipeline near-linear ($O(n \log n)$) through adaptive partition splitting, scaling to 3M-sample inputs; wrote the core geometry kernels (sphere exposed-region clipping, multi-level spatial hashing for sphere intersection) in C++ with pybind11.

Inheritance analysis based on 3D skeletonization | Mathematica,

St. Louis, MO

Python

Guided by Professor Tao Ju, Washington University in St. Louis

Dec. 2022 - May 2023

- **Feature Extraction:** Analyzed 3D spatial density data of chromosomes and extracted skeletal features such as chromosome length, inter-chromosomal angles, and dimensions along three PCA directions.
- **Threshold Optimization:** Computed skeletal topological change curves for each sample across different density thresholds, by which I implemented an algorithm to automatically determine the optimal thresholds for binarization and skeletonization.
- **Dimensional Reduction & Data Analysis:** Calculated over 50 features from each sample's skeleton, performed dimensionality reduction methods (PCA, UMAP, and LDA), and identified patterns and trends to locate affected genes.

AWARDS & HONORS

Summer PhD Research Award, George Mason University

2026

Contemporary Undergraduate Mathematical Contest in Modelling

Sept. 2019

The National 2nd Prize, Ranking: top 3.8%